

1700-1799

INDUSTRIALIZATION

In the 18th century, technological innovation and widespread societal changes collided to transform Britain's economy. Better agricultural methods made farms more productive, creating a ready supply of labour, while new banking practices freed up capital for investment. Steam engines replaced horses and water as the source of power for factories and transport, and allowed mines to efficiently produce the raw materials for Britain's expanding iron industry. The iron industry was transformed by the shift from charcoal to coal furnaces in 1709, and new puddling and rolling techniques for removing impurities from iron were developed in 1783, unlocking large-scale iron production.

Inventions such as the spinning jenny and Arkwright's water frame took the textile industry out of artisans' cottages and into giant factories. As iron foundries and textile mills moved near to Britain's coalfields in the Midlands and north of England and South Wales, a network of canals and roads sprang up to carry goods across the country.

Industrialization gave Britain an economic advantage that translated into political power, and also brought social change as workers came together, often striking or destroying machinery, in the fight for better pay and working conditions. By the 19th century, Germany, France, the US, and Japan had joined the Industrial Revolution, transforming the modern world.

KEY MOMENTS

1712 Newcomen's engine transforms mining

Coal was the fuel of the Industrial Revolution, key to the production of iron and steel. Its mining was transformed in 1712 when Thomas Newcomen developed his atmospheric steam engine, which cheaply and efficiently pumped water from Britain's mines, allowing them to be exploited to ever greater depths. By 1729, over 100 engines were at work in Britain and across Europe. The vast amounts of coal so produced transformed the industrial landscape (*right*).

1779 World's first iron bridge constructed

Built over the River Severn in Shropshire, England, the Iron Bridge (*right*) was a turning point in engineering, demonstrating what could be achieved with cast iron - the material that defined the new industrial age. Demand for iron for steam engines and military hardware in the Napoleonic Wars meant that Britain's iron production quadrupled between 1793 and 1815.

